

Analysis Service Working

1. What We Send to Gemini API:

We send a structured summary, not raw code. Here's the exact flow:

1. Data Collection (Your Code Already Does This)

Project Analysis Results:

- Project Type: JAVA
- Main Language: Java
- Build Tool: Maven
- Architecture: FULL_STACK
- Key Files: [pom.xml, src/main/, README.md]
- Dependencies: [Java project - check pom.xml]
- File Structure: {Map of file paths → "FILE" or "DIR"}
- Sub-projects: [backend (JAVA), frontend (NODE_JS)]

2. What Gets Sent to Gemini:

Project Details:

- Name: MySpringApp
- Type: JAVA (Full-stack)
- Main Files: pom.xml, src/main/java/, src/components/
- Key Directories: controllers/, services/, models/, public/
- Build: Maven
- Has: Backend (Java/Spring), Frontend (React components)
- Special Files: Dockerfile, application.properties

3. How We Read Project Files:

```
private Map<String, String> scanProjectStructure(String localPath) throws IOException {  
    // This ONLY collects FILE NAMES and PATHS, not file content!
```

```
// Example output:

// {

//  "src/main/java/com/example/Main.java": "FILE",
//  "src/components/Header.js": "FILE",
//  "pom.xml": "FILE",
//  "src/": "DIR"

// }

// No file content read here!

}
```

Why This Works:

1. **Token Limit Safe:** We send ~100-500 words, not thousands of lines of code
2. **AI Doesn't Need Code:** Gemini only needs to understand the **project structure** and **technology stack** to generate a good README
3. **Smart Filtering:** We only mention key/important files, not every single file

Complete Example - What Actually Goes to Gemini:

The prompt sent to Gemini looks like:

String prompt = ""

Project Details:

Project Name: E-commerce Backend

Repository: github.com/example/ecom

Template Domain: WEB_SOFTWARE

Project Type: JAVA

Architecture: BACKEND_ONLY

Main Language: Java

Build Tool: Maven

Key Files: [pom.xml, src/main/java/, application.properties, Dockerfile]

Sub-projects: []

Dependencies: [Spring Boot, Hibernate, PostgreSQL]

File Structure: Contains controllers/, services/, models/, repository/ directories

""",

Key Insight:

- **File scanning** (Files.walk) → Gets **file paths only** (not content)
- **Analysis** → Determines project type/structure from **file presence**
- **Prompt building** → Creates a **textual summary** of findings
- **Gemini receives** → Only the summary, understands the "shape" of the project
- **Output** → Context-aware README based on project type/structure

What We NEVER Send:

- Actual source code content
- Entire file contents
- Binary files or libraries
- All files in node_modules/

The scanProjectStructure() returns:

```
{
    "src": "DIR",
    "src/main": "DIR",
    "src/main/java": "DIR",
    "src/main/java/com": "DIR",
    "src/main/java/com/example": "DIR",
    "src/main/java/com/example/dto": "DIR",
    "src/main/java/com/example/dto/UserDTO.java": "FILE",
    "src/main/java/com/example/dto/ProductDTO.java": "FILE",
    "src/main/java/com/example/dto/OrderDTO.java": "FILE",
    "src/main/java/com/example/controller": "DIR",
    "src/main/java/com/example/controller/UserController.java": "FILE",
    "src/main/java/com/example/controller/ProductController.java": "FILE"
}
```

Even Smarter: identifyKeyFiles()

Another function filters this further:

```
private List<String> identifyKeyFiles(String localPath) {
    (Only checks for IMPORTANT files, not all files)
    String[] importantFiles = {
        "README.md", "package.json", "pom.xml", "build.gradle",
        "requirements.txt", "Dockerfile", "docker-compose.yml",
        ".env", "src/", "main.py", "app.js", "index.js", "Main.java",
        "application.properties", "config/", "public/", "src/main/", "src/test/"
    };
    (Returns: ["pom.xml", "src/main/", "application.properties"])
}
```

Final Gemini Prompt Looks Like:

Project has:

- dto/ directory with multiple DTO classes
- controller/ directory with REST controllers
- src/main/java/ structure
- pom.xml file
- application.properties

This indicates a Java Spring Boot application with:

- Data Transfer Objects for API requests/responses
- REST controllers for HTTP endpoints
- Maven build system
- Configuration via properties files

Why This Works So Well

1. AI Training Data

Gemini has been trained on **millions** of GitHub projects. When it sees:

- pom.xml + src/main/java/ + controller/ + dto/
→ It **immediately knows** this is a Spring Boot REST API

2. Common Patterns Recognition

For a Java project, the README **doesn't need** to know your specific business logic. It needs:

- **Setup instructions** (mvn install, mvn spring-boot:run)
- **Project structure** explanation
- **API documentation** format
- **Database configuration**
- **Testing commands**

3. Real-World Example:

Your project structure tells Gemini:

Input: Java + Maven + controllers/ + dto/ + application.properties

↓

AI's Understanding: "Spring Boot REST API with DTOs and controllers"

↓

Output README includes:

- Prerequisites: Java 17, Maven, PostgreSQL
- Running: `mvn spring-boot:run`
- API endpoints section
- DTO structure explanation
- Database setup

4. What Gemini INFERS (Doesn't Need Code):

From dto/UserDTO.java **filename alone**, Gemini knows:

- This is a Data Transfer Object

- It's probably used in API requests/responses
- Should be mentioned in the "API Structure" section
- Follows Java naming conventions

The Key Insight:

READMEs are 80% boilerplate + 20% project-specific. The AI needs to know:

1. **Technology stack** (Java/Spring vs Node/Express vs Python/Django)
2. **Project type** (API vs Web App vs Library)
3. **Architecture** (MVC, Microservices, etc.)
4. **Key files** that matter for setup

Your file/folder names give it **exactly that!**

Diff Analysis

What It DOES Analyze:

// Output from this service:

```
GitDiffAnalysisDTO {  
    filesChanged: 3,           // Number of files changed  
    changeSummary: "Added: UserService.java, Modified: README.md, Deleted: OldFile.txt",  
    modifiedFiles: ["src/UserService.java", "README.md"], // List of file paths  
    commitHash: "abc123..."  
}
```

Working

Step 1: Get Current & Previous Commits

```
ObjectId head = repository.resolve("HEAD"); // Current commit  
RevCommit currentCommit = revWalk.parseCommit(head); // Parse it  
RevCommit parentCommit = currentCommit.getParent(0); // Previous commit
```

Step 2: Compare Two Git Trees

```
// Get file trees from both commits  
Tree oldTree = prepareTreeParser(repository, previousHead); // Old state  
Tree newTree = prepareTreeParser(repository, head); // New state  
  
// Compare: what changed between these two states?  
List<DiffEntry> diffs = git.diff()  
    .setOldTree(oldTree)  
    .setNewTree(newTree)  
    .call();
```


Step 3: Extract Metadata from Diffs

Each DiffEntry contains:

```
diff.getChangeType() // ADD, MODIFY, DELETE, RENAME, COPY
diff.getNewPath()    // File path (e.g., "src/main/java/Service.java")
diff.getOldPath()    // Old path (for renames)
```

Step 4: Create Summary

// Example output:

"Added: UserService.java, Modified: README.md, Deleted: config.ini"

What GitDiffService.analyzeChanges() Actually Does:

```
public GitDiffAnalysisDTO analyzeChanges(ProjectEntity project, String commitHash) {
    // 1. Open the Git repository
    Repository repository = builder.setGitDir(new File(project.getLocalPath(), ".git")).build();

    // 2. Get CURRENT commit (HEAD)
    ObjectId head = repository.resolve("HEAD"); // Current commit

    // 3. Get PREVIOUS commit (HEAD~1)
    RevWalk revWalk = new RevWalk(repository);
    RevCommit currentCommit = revWalk.parseCommit(head);

    // Check if there's a parent (not first commit)
    if (currentCommit.getParentCount() > 0) {
        RevCommit parentCommit = currentCommit.getParent(0); // Previous commit
        ObjectId previousHead = parentCommit.getId();        // Parent commit ID
    }
}
```

```
// 4. Compare PREVIOUS vs CURRENT
```

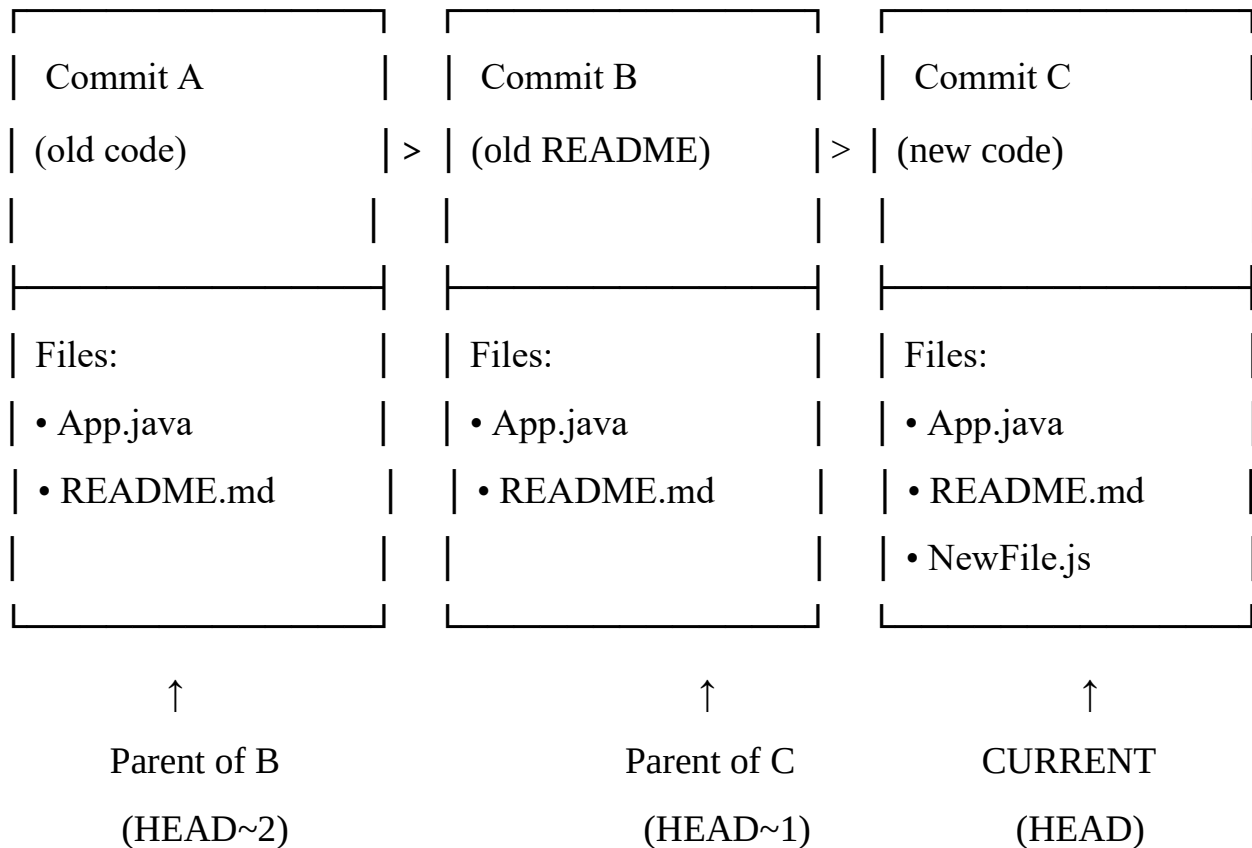
```
List<DiffEntry> diffs = git.diff()

    .setOldTree(prepareTreeParser(repository, previousHead)) // OLD state
    .setNewTree(prepareTreeParser(repository, head))          // NEW state
    .call();
}
```

VISUAL REPRESENTATION:

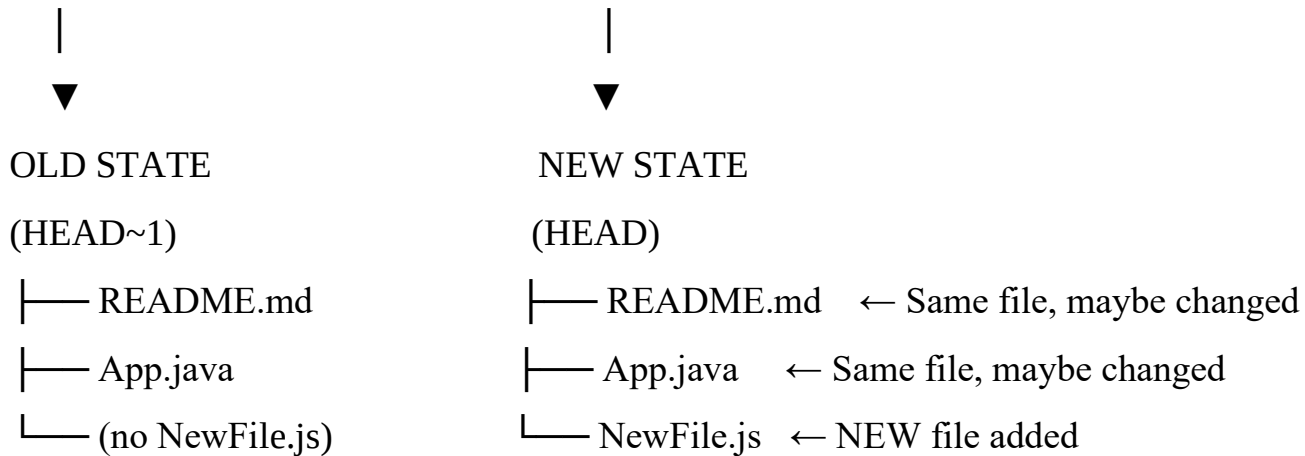
Git Commit Timeline:

Commit History:



What Gets Compared:

COMPARISON: [Commit B] vs [Commit C]

**CODE BREAKDOWN: The Parent Commit Logic**

// Get CURRENT commit (latest)

RevCommit currentCommit = revWalk.parseCommit(head);

// currentCommit = Commit C from diagram

// Get its PARENT (previous commit)

if (currentCommit.getParentCount() > 0) {

RevCommit parentCommit = currentCommit.getParent(0);

// parentCommit = Commit B from diagram

// Now we can compare B vs C

// B = "old state" (before latest commit)

// C = "new state" (after latest commit)

}

// Why check parent count?

// First commit has no parent! Can't compare.

RELATION TO README GENERATION:

The Logic Flow:

1. **Git Hook fires** after git commit
2. **Backend receives** the new commit hash
3. **GitDiffService compares:**
 - HEAD~1 (state before commit) ← **OLD README here**
 - HEAD (state after commit) ← **NEW CODE here**
4. **AI Service gets:**
 - Old README content (from database, not from Git)
 - List of changed files (from Git diff)
5. **AI generates** new README based on code changes